

BUILDING WITH BLUEPRINTS

THE QUIZ MASTER | SESSION 1

Coding is like magic—you build your own world with simple blueprints! Today, we are moving beyond simple lines of code to become master architects of our own Quiz Game.

The Challenge of Manual Quizzing

Imagine trying to manage 100 questions one by one. Without a plan, your code becomes messy and hard to track. It's easy to lose track of which questions you've already asked!

The Question: What is the national fruit of India?

The Answer: Mango

Now imagine writing this code 100 times. There must be a smarter way!

Class vs. Object: The Blueprint

In Python, we use something called a **Class** to solve our messy code problem.

THE RUBBER STAMP ANALOGY

Think of a Class like a rubber stamp design. You create the design once. Every time you press that stamp on paper, you get a real ink mark—this is the **Object**!

One Stamp Design = Many Ink Marks!

- **CLASS = BLUEPRINT:** The idea or design template.
- **OBJECT = REAL THING:** The actual thing created from the template.

Our 'Question' Blueprint

Let's write our first blueprint in Python code. This will be the design for every quiz question we ever make!

THE BLUEPRINT CODE

```
class Question:
    def __init__(self, text, answer):
        self.text = text
        self.answer = answer
```

WHAT DOES THIS DO?

- **Class Question:** This tells Python we are making a new blueprint.
- **The `__init__` Method:** This is a special method that sets up each object's information.
- **Attributes:** These are the pieces of information each object holds—like the question text and its answer!

Creating Real Questions

Now that we have our stamp (the Class), let's make some ink marks (the Objects)!

CODE EXAMPLE

```
q1 = Question("Capital of India?", "New Delhi")
q2 = Question("IPL 2023 winner?", "CSK")
q3 = Question("ISRO chief?", "Dr. S. Somanath")
```

STORING OUR OBJECTS

We can put all our "real things" into a list so they are easy to manage.

```
questions = [q1, q2, q3]
```

Now we have 3 questions ready to play!

How the Quiz Game Works

To run our game, we need to go through our list and do three things for every question:

1. **Show Question Text:** Display the question on the screen so the player can read it.
2. **Get Answer:** Wait for the player to type their answer using the `input()` function.
3. **Check & Respond:** Compare the player's answer to the real answer and say "Correct!" or "Try again!"

Code Tip: We use a `for` loop to go through our list of question objects one by one!

The Quiz Logic

Here is how we check the answers using Python code:

```
for question in questions:
    user_answer = input(question.text + " ")
    if user_answer.lower() == question.answer.lower():
        print("Correct!")
    else:
        print("Incorrect. Try next question.")
```

INDIAN GK CHALLENGE!

Can you answer these questions in your game?

- **Q2:** Which city is called the Silicon Valley of India? (Answer: Bengaluru)
- **Q3:** Which Indian state is famous for the Sun Temple? (Answer: Odisha)

What Did We Learn Today?

- A Class is a blueprint for creating objects.
- An Object is the real thing made from that blueprint!
- The `__init__` method sets up each object's information.
- We built a quiz game using question objects!

WHAT'S NEXT?

Get ready for Session 2! We will learn how to track your points, add levels, and create a real quiz competition!

Coming Soon!

"Coding is the power to bring your ideas to life!"

Thank you for coding with us!

Planrs Academy